IC4SD-TrafficSnap: A City-Scale Multi-Camera Image Time-Series and Geospatial Road Network Dataset for Heterogeneous Urban Traffic

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Abstract

This data article presents **IC4SD-TrafficSnap**, a multimodal dataset coupling city-scale surveillance camera image time-series and geospatial road network mapping across **608** traffic surveillance stations in the urban and peri-urban corridors of Ho Chi Minh City, Vietnam. The visual corpus contains **714,123** discrete camera snapshots (**44.38 GiB** / **47.66 GB** in JPEG format) captured over **93.6 continuous hours** (October 2–6, 2026). Across active acquisition sessions ($\Delta T < 3,600$ s), the pairwise inter-snapshot interval averages $\mathbf{269.0 \pm 239.7}$ s (median: 263.0 s, 88.4% $\leq$ 300 s, $p_{90} \approx 312.0$ s), corresponding to a station-wise wall-clock interval of 286.9 s/frame and a fleet-wide aggregate throughput of 0.47 s/frame across all 608 stations. Physical shortest-path driving distances and directions are extracted from OpenStreetMap using the Open Source Routing Machine (OSRM v5.27.1). From these geospatial road assets, a 608×608 directed spatial graph containing **2,450** valid corridor links within a **6.0 km** search cutoff horizon is derived as a physical road connectivity representation, capturing directional asymmetry (**1,070** unidirectional links and **238** distance-asymmetric bidirectional pairs, alongside **452** metric-symmetric pairs). PyTorch data loading toolchains and dual-transition directed diffusion operators are provided. Privacy assessments combining automated detector pre-screening across the full census and targeted multi-rater manual inspection of 1,000 worst-case daylight frames confirmed zero readable license plates or identifiable facial biometrics, establishing a 95% confidence upper bound of $p \leq 0.30\%$ on that targeted sample via the Rule of Three. The dataset is released under a three-tier licensing framework: CC BY-NC 4.0 for imagery, ODbL for the road graph, and MIT for software utilities.

Keywords: Intelligent Transportation Systems, Image Time-Series, Geospatial Road Network, Spatial Graph Topology, Mixed Urban Traffic

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Table 1: Specifications Table of the IC4SD-TrafficSnap Dataset.

Subject	Computer Science
Specific subject area	Intelligent Transportation Systems, Computer Vision, Spatio-Temporal Data Mining
Type of data	Image time-series, geospatial metadata, derived road network graph tensors, PyTorch code utilities
How data were acquired	Recorded from open public municipal traffic monitoring camera streams operated under the intelligent transportation system (ITS) of the Ho Chi Minh City Department of Transportation (`https://giaothong.hochiminhcity.gov.vn`) for civic traffic monitoring and open public research. Driving distances and road network routing were computed using OSRM over OpenStreetMap data.
Data format	Raw snapshots: RGB JPEG (512×288 px, JFIF 1.01, stream quality factor $\approx$ 75–80); Geospatial metadata: Tabular CSV (`routes.csv`, `stations.csv`, `road_network_distance.csv`); Graph tensors: NumPy binary arrays (`distance_km.npy`, `direction.npy`) and tabular edge list (`edges.csv`); Code: Python source files (`code/`).
Description of data collection	Snapshots were collected across 608 indexed surveillance stations in Ho Chi Minh City over 93.6 continuous hours (October 2, 2026, 17:37 to October 6, 2026, 15:12 ICT) with an empirical pairwise active sampling interval of $\Delta T = 269.0 \pm 239.7$ s (median: 263.0 s, 88.4% $\leq$ 300 s, $p_{90} \approx 312.0$ s, $p_{95} \approx 520.0$ s, $p_{99} \approx 1,200.0$ s; station-wise wall-clock interval: 286.9 s/frame/station; fleet aggregate throughput: 0.47 s/frame; global interval range: [2.0 s, 4.12 h] reflecting distributed upstream network latency alongside prolonged link outages at 14 peripheral stations). Camera positions are mapped to OpenStreetMap road geometry via a three-stage sequential corridor pruning pipeline, yielding a derived directed spatial graph with 2,450 corridor links within a 6.0 km driving search horizon (1,070 unidirectional links and 238 distance-asymmetric pairs, alongside 452 metric-symmetric pairs; network tortuosity $\tau = 1.25 \pm 0.61$).
Data source location	Region: Ho Chi Minh City, Vietnam (metropolitan urban core and peri-urban corridors including Thu Duc City, District 12, Binh Chanh, and Hoc Mon); Coordinates: [10.642472°N − 10.988295°N, 106.452713°E − 106.850624°E] (dense urban core: [10.70°N − 10.88°N, 106.60°E − 106.78°E]).
Data accessibility	Repository name: **Zenodo** Data identification number (DOI): **10.5281/zenodo.22929940** Direct URL to data: `https://doi.org/10.5281/zenodo.22929940` Licensing: • Visual surveillance snapshots: *Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0)* • Derived road network graph and distance tensors: *Open Database License 1.0 (ODbL)* • Software utilities and loaders: *MIT License*
Related research article	V.-A. Le, D. Hoang-Viet, K. Nguyen-Trong, "Urban Traffic Perception and Multi-Horizon Graph Forecasting: A Two-Stage Camera-Sensed Framework", *arXiv preprint arXiv:2602.04159* (2026). Companion pilot benchmark dataset (v1.0): Zenodo, DOI: `https://doi.org/10.5281/zenodo.22089673`.

1. Value of the Data

- **Heterogeneous Mixed Urban Traffic Representation:** Captures multi-day visual dynamics of a Southeast Asian megacity where two-wheeled motorized vehicles account for **70–80%** of modal traffic volume [1]. This fills a critical empirical gap in intelligent transportation benchmarks, supporting macroscopic traffic state analysis and scene representation under dense mutual occlusions and non-lane-based movement patterns.

- **Macroscopic Spatial-Temporal Vision vs. Microscopic Tracking:** Native 512×288 resolution is deliberately matched with a 5-minute sampling cadence to enable city-scale macroscopic traffic state estimation, roadway background decomposition, and self-supervised visual representation pre-training (ViT, DINOv2, MAE) across 608 endpoints without prohibitive network bandwidth overhead or privacy risks.

- **Geospatial Road Network and Derived Directed Graph Topology:** Integrates surveillance visual streams with OpenStreetMap coordinates and shortest-path driving distance matrices (608×608, **2,450** valid corridors). The directed spatial graph serves as a derived topological representation of physical road infrastructure connectivity (featuring 1,070 unidirectional links and 238 distance-asymmetric bidirectional pairs, alongside 452 metric-symmetric pairs; network tortuosity $\tau = 1.25 \pm 0.61$), providing a formal graph-theoretic substrate for structural network analysis, spatial corridor modeling, and geometric deep learning on urban transportation infrastructure.

- **Roadway Background and Incident Modeling:** The fixed-viewpoint multi-camera time-series across circadian day-night cycles provides an empirical basis for separating static roadway backgrounds from transient vehicular flow, enabling unsupervised anomaly detection, flood monitoring, and environmental scene decomposition.

- **Open Reproducibility and Three-Tier Licensing:** Accompanied by verified PyTorch data loaders, multi-camera batch samplers, and dual-transition directed graph processing routines. The dataset operates under a transparent three-tier licensing framework: visual surveillance imagery is released under *Creative Commons Attribution-NonCommercial 4.0 (CC BY-NC 4.0)* for academic research, the derived road network graph is released under the *Open Database License (ODbL)*, and software utilities are distributed under the *MIT License*.

2. Background

Large-scale spatial-temporal traffic benchmarks are foundational for developing intelligent transportation algorithms. However, established benchmarks in literature diverge into three disconnected paradigms:

1. **Longitudinal 1D sensor-based traffic datasets** (such as METR-LA [2], PEMS-BAY [2], and LargeST [3]) provide extensive graph topologies and longitudinal 1D time-series, but lack visual context, preventing models from leveraging rich photometric, meteorological, and vehicle-mix features.

2. **Ego-vehicle driving benchmarks** (e.g., BDD100K [4]) provide rich visual sequences from moving dashcams, but capture singular mobile viewpoints lacking static city-scale surveillance infrastructure and continuous multi-station network topologies.
3. **Localized computer vision surveillance benchmarks** (e.g., UA-DETRAC [5], CityFlow [6], MIO-TCD [7], and CityCam [8]) offer multi-camera visual imagery, but either focus on microscopic vehicle tracking across a handful of isolated intersections (24–40 cameras) or provide static disconnected image crops lacking continuous time-series and formal city-scale geospatial routing tensors. Furthermore, they predominantly reflect car-dominant traffic in temperate North American or East Asian settings.

IC4SD-TrafficSnap bridges this fundamental division by providing an integrated multimodal dataset: continuous surveillance camera snapshots and geospatial road network mapping across **608** camera stations in Ho Chi Minh City, Vietnam. From these primary road mapping and spatial assets, a directed spatial graph topology is established as a direct derived representation to support spatial graph representation learning and macroscopic transportation analysis. A systematic comparison with existing benchmarks is summarized in Table 2.

Table 2: Systematic comparison of IC4SD-TrafficSnap with established benchmarks in intelligent transportation systems and computer vision.

Dataset	Urban Context & Flow	Nodes	Time Span	Sensing Modality & Topology	Primary Target Research Scope
A. Longitudinal 1D Sensor Infrastructure Benchmarks					
METR-LA [2]	Los Angeles, USA (Car-dominant)	207	4 months	Inductive loop sensors (Directed distance graph)	1D traffic speed forecasting (lacks visual context)
PEMS-BAY [2]	San Francisco, USA (Car-dominant)	325	6 months	Inductive loop sensors (Directed distance graph)	1D traffic speed forecasting (lacks visual context)
LargeST [3]	California, USA (Car-dominant)	8,600	5 years	Inductive loop sensors (Large-scale highway graph)	Macroscopic network forecasting (lacks visual context)
B. Ego-Vehicle / Localized Microscopic Video Benchmarks					
BDD100K [4]	United States (Car-dominant)	Mobile	100K clips	Ego-vehicle dashcam video (No fixed surveillance graph)	Autonomous driving multitask perception (lane, drivable, tracking)
UA-DETRAC [5]	Beijing / Tianjin, China (Car-dominant)	24	Short clips	960×540 surveillance video (Isolated intersections, no graph)	Microscopic vehicle detection and multi-object tracking
CityFlow [6]	Mid-sized city, USA (Car-dominant)	40	3.25 hours	HD synchronized video clips (Small intersection cluster, no tensor)	Multi-target multi-camera tracking (MTMC) and vehicle Re-ID
MIO-TCD [7]	North America (Car-dominant)	Multiple (unindexed)	Discontinuous	Isolated crop snapshots (Unconnected cameras, no graph)	Vehicle categorization and localization (no continuous dynamics)
CityCam [8]	New York City, USA (Car-dominant)	212	Multi-day	Low-resolution webcams (No formal routing graph)	Vehicle counting and density estimation across disparate viewpoints
C. City-Scale Multimodal Spatial-Temporal Vision Benchmark					
IC4SD-TrafficSnap	**Ho Chi Minh City, VN (Motorcycles: 70–80%)**	**608**	**93.6 hours**	**Multimodal:** 512×288 **RGB snapshots + Geospatial road network (2,450 directed corridors)**	**Self-supervised visual representation, roadway background modeling, and macroscopic urban traffic scene understanding**

3. Data Description

3.1. Repository Organization, Lineage, and Archive Structure

The dataset is permanently hosted on Zenodo (`https://doi.org/10.5281/zenodo.22929940`, Dataset Release v2.0).

Dataset Lineage and Version Evolution: The IC4SD data initiative originated with an initial pilot benchmark (*IC4SD-Traffic-HCM*, v1.0, Zenodo DOI: `https://doi.org/10.5281/zenodo.22089673`), comprising 5,012 annotated frames and historical flow aggregates across 86 days for supervised vehicle detection and preliminary graph exploration (2,452 candidate edges) [9]. **IC4SD-TrafficSnap** represents a major city-scale expansion (Release v2.0), scaling the raw visual corpus by over 140× to **714,123 continuous snapshots** across 93.6 continuous hours and 608 stations. The road network topology has been rigorously standardized into a 608×608 directed spatial graph with exactly **2,450 valid corridor links** (after resolving 2 redundant multi-mast duplicate loop links at complex multi-way junctions), accompanied by dual directed random-walk transition operators and full empirical photometric and physical privacy audit protocols. The repository is organized into modular directories as illustrated in Figure 1:

3.2. Data Organization and Comprehensive Data Dictionary

The repository archive is organized hierarchically as shown in Figure 1. To ensure transparent documentation and facilitate immediate downstream adoption, a comprehensive data dictionary specifying file formats, tensor shapes, variable definitions, and concrete sample records across all released assets is detailed in Table 3.

Table 3: Comprehensive Data Dictionary: Multi-Part Schemas, Data Types, and Concrete Sample Records.

Asset / File Path	Shape / Volume	Data Format	Exact Schema & Dtypes	Description & Concrete Sample Record
Part A: Tabular Geospatial and Network Metadata CSV Assets				
`metadata/routes.csv`	608×8 tabular records	UTF-8 text CSV	(1) `matrix_index` (int64) (2) `stt` (int64) (3) `Location` (string) (4) `CamID` (string, hex-24) (5) `latitude` (float64) (6) `longitude` (float64) (7) `coord_pair_str` (string) *[col: "latitude, longitude"]* (8) `Phuong` (string)	Canonical camera registry containing exactly 8 columns synchronizing dense 0-indexed matrix indices with official hexadecimal stream identifiers and administrative wards across HCMC. Note: Column 7 contains the literal CSV header `"latitude, longitude"` storing the combined coordinate string. Camera elevations ($H \in [6.0, 15.0]$ m) are cataloged from municipal ITS engineering specifications. *Exact 8-element sample row:* `0, 1, "Tran Quang Khai - Tran Khac Chan", "662b86c41afb9c00172dd31c", 10.791890, 106.691054, "10.791890, 106.691054", "Phuong Tan Dinh, Quan 1"`
`metadata/stations.csv`	608×5 tabular records	UTF-8 text CSV	(1) `matrix_index` (int64) (2) `station_id` (int64) (3) `out_degree` (int64) (4) `in_degree` (int64) (5) `node_type` (string)	Station-level topological degree catalog and structural role classification within the directed road network graph (`regular`, `source`, `sink`, `isolated`). *Exact 5-element sample row:* `0, 1, 7, 5, "regular"` (Station 1 has out-degree 7, in-degree 5, classified as regular transit node).
`graph/edges.csv`	$2,450 \times 9$ tabular links	UTF-8 text CSV	(1) `source_station_id` (int64) (2) `target_station_id` (int64) (3) `source_idx` (int64) (4) `target_idx` (int64) (5) `distance_km` (float64) (6) `distance_m` (float64) (7) `is_bidirectional` (bool) (8) `reverse_dist_km` (float64) (9) `dist_diff_m` (float64)	Exhaustive edge list of all 2,450 valid directed arterial corridors extracted within the 6.0 km OSRM navigability horizon. Unidirectional links record `NaN` for reverse metrics. *Exact 9-element sample row:* `1, 7, 0, 5, 1.74, 1740.0, False, NaN, NaN` (directed corridor from Station 1 [`idx=0`] to Station 7 [`idx=5`] of length 1.74 km without direct reverse edge).
Part B: Machine-Learning Ready Graph Tensors				
`graph/distance_km.npy`	Matrix tensor: $(608, 608)$ 2D	NumPy binary (`float64`, little-endian, 2.82 MB)	$d_{ii} = 0.0$ (self) $d_{ij} \in (0.0, 6.0]$ km (valid) $d_{ij} = \text{NaN}$ (unconnected)	Directed driving distance tensor along OpenStreetMap corridors. Self-distances along diagonal are 0.0. Valid corridors (2,450 links) contain driving distances in kilometers. Unconnected pairs (366,606 entries, 99.17%) are explicitly encoded as IEEE 754 floating-point `NaN`. When loaded via `code/graph_utils.py`, NaNs are mapped to 0.0 via `np.nan_to_num`. *Sample slice:* `dist[0, 5] = 1.74`; `dist[0, 1] = NaN`.
`graph/direction.npy`	Matrix tensor: $(608, 608)$ 2D	NumPy binary (`int8`, little-endian, 361 KB)	$A_{ij} \in \{0, 1\}$ (1 = link, 0 = none)	Directed binary reachability adjacency matrix (1 denotes a verified navigable directed road corridor from station i to station j; 0 denotes absence of direct link or self-loop). Non-zero entries count: 2,450.
Part C: Visual Surveillance Image Corpus				
`images/station_<id>/*.jpg`	714,123 image files (44.38 GiB /	RGB JPEG image (JFIF 1.01	$512 \times 288 \times 3$ px 66.74 KB (65.18 KiB) $Q \approx 75$–80	Surveillance snapshots partitioned into per-station subdirectories, following the deterministic pattern `station_<id>`

IC4SD-TrafficSnap: Zenodo Deposition Archive Directory Structure

```
IC4SD-TrafficSnap/ ..... Root open-access repository archive
├── LICENSE ..... Three-tier licensing (CC BY-NC 4.0, ODbL, MIT)
├── CITATION.cff ..... Citation File Format metadata (v1.2.0)
├── .zenodo.json ..... Zenodo persistent DOI repository metadata schema
├── data_dictionary.md ..... Comprehensive schema and column attribute definitions
├── README.md ..... Usage documentation and quick-start verification
├── images/ ..... Full corpus of 714,123 JPEG snapshots (44.38 GiB / 47.66 GB)
│   ├── station_1/ ..... Continuous snapshots: station 1 timestamp sequence
│   ├── ..... Sharded by station ID across all 608 active endpoints
│   └── station_657/
├── metadata/ ..... City-scale geospatial coordinates and routing attributes
│   ├── routes.csv ..... Canonical camera registry: 608 stations, CamIDs, coordinates
│   ├── road_network_distance.csv ..... Directed OSM shortest driving distance matrix
│   │   (608x608)
│   └── stations.csv ..... Station mapping, in/out-degrees and topological role registry
├── graph/ ..... Machine-learning ready graph tensors and corridor lists
│   ├── distance_km.npy ..... Routing distance tensor in kilometers (608x608, float64,
│   │   NaN=unconnected)
│   ├── direction.npy .. Binary directed adjacency matrix (608x608, int8, 1=edge, 0=none)
│   └── edges.csv ..... Tabular list of 2,450 directed corridor links (d <= 6.0 km)
├── sample_preview/ ..... Starter subset for rapid environment verification (500 MB)
│   └── sample_camera_sequences/ ..... Representative snapshots across sample stations
└── code/ ..... PyTorch data loading and graph modeling utilities
    ├── load_unlabeled_images.py ... TrafficCameraUnlabeledDataset PyTorch loader and
    │   temporal indexing
    ├── camera_sampler.py ..... Multi-camera SSL batch sampler (DINO/MAE-ready)
    └── graph_utils.py ..... Dual directed random walk operators (Pf, Pb) and Chebyshev
        Laplacian
```

Figure 1: Modular directory structure and asset organization of the IC4SD-TrafficSnap Zenodo repository archive. In the permanent repository, the visual corpus of 714,123 JPEG snapshots is provided as partition archives expanding into the `images/station_<id>/` directory tree, alongside a lightweight `sample_preview/` bundle (≈ 500 MB) for immediate zero-download inspection.

3.3. Visual Snapshot Modality and Temporal Sampling

The visual corpus consists of discrete 3-channel RGB JPEG snapshots (JFIF 1.01 standard) with a uniform native frame resolution of $\mathbf{512} \times \mathbf{288}$ pixels (standard stream compression quality factor ≈ 75–80). The total visual archive occupies **44.38** GiB in binary gibibytes (1 GiB = 2^{30} bytes = 1,073,741,824 bytes), corresponding to **47.66** GB in standard decimal gigabytes (1 GB = 10^9 bytes). The mean snapshot file size is $\mathbf{66.74} \pm \mathbf{13.68}$ KB in decimal kilobytes, which mathematically evaluates to exactly $\mathbf{65.18} \pm \mathbf{13.36}$ KiB in binary kibibytes ($714,123 \times 65.18$ KiB ≈ 44.39 GiB, equivalent to 47.66 GB decimal), ensuring complete numerical consistency between individual file measurements and overall disk volume. Each image file adheres to a deterministic, machine-readable filename schema:

$$\text{Filename} = \texttt{<station_id>_<unix_timestamp>.jpg} \tag{1}$$

where `station_id` represents the historical integer station identifier ($1 \leq$ `station_id` ≤ 657, indexing the 608 active stations cataloged across the municipal network and preserved from the initial pilot benchmark *IC4SD-Traffic-HCM* v1.0), the underscore is a delimiter, and `unix_timestamp` represents the 10-digit integer UTC epoch seconds recorded upon frame arrival (Vietnam Standard Time is UTC + 7). An explicit structural breakdown of a representative image file name is provided in Table 4.

Table 4: Structural breakdown and semantic interpretation of canonical image filename `1_1759401420.jpg`.

Token	**Data Type**	**Example Value**	**Semantic Interpretation & Practical Context**
`station_id`	Integer	`1`	Historical station identifier ($1 \leq$ ID ≤ 657), indexing Station 1 at Tran Quang Khai – Tran Khac Chan.
`_`	Delimiter	`_`	Underscore delimiter separating station index and timestamp.
`unix_timestamp`	Integer	`1759401420`	UTC epoch seconds ($1,759,401,420 \equiv$ 2026-10-02 17:37:00 ICT, UTC+7).
`.jpg`	Extension	`.jpg`	3-channel RGB image encoded in baseline JFIF 1.01 JPEG format.

Tripartite Station Identifier Mapping Architecture. To ensure persistent interoperability across raw image filenames, graph matrices, and municipal open data infrastructure, each surveillance endpoint is cross-referenced through a unified tripartite schema:

1. **Continuous Matrix Index ($0 \leq$ `matrix_index` ≤ 607):** A dense 0-indexed integer defining the row and column coordinate of the station in the 608×608 spatial graph adjacency and distance tensors (`graph/distance_km.npy`, `graph/direction.npy`).
2. **Historical Station Identifier ($1 \leq$ `station_id` ≤ 657):** The legacy station identifier (`stt`) retained from the initial pilot benchmark (*IC4SD-Traffic-HCM* v1.0) and embedded directly into image filenames ($\langle$`station_id`$\rangle$ `_` $\langle$`timestamp`$\rangle$`.jpg`). Exactly 608 active endpoints are indexed across the range $[1, 657]$, with legacy gaps corresponding to inactive or decommissioned municipal feeds.
3. **Municipal Technical Identifier (`CamID`):** The canonical 24-character hexadecimal camera ID assigned by the Ho Chi Minh City Department of Transportation streaming registry (`https://giaothong.hochiminhcity.gov.vn`), ensuring persistent traceability to official municipal camera metadata.

A complete cross-reference lookup table synchronizing `matrix_index`, `station_id`, `CamID`, arterial locations, and geographic coordinates is archived in `metadata/routes.csv` and `metadata/stations.csv`. Exactly **608 camera stations** are indexed in the dataset across the municipal road network.

Temporal Sampling Dynamics and Empirical Fleet Cadence. Snapshots were recorded across **93.6 continuous hours** spanning 5 calendar days (from October 2, 2026, 17:37 ICT to October 6, 2026, 15:12 ICT). The empirical inter-snapshot temporal dynamics exhibit two complementary statistical perspectives:

- **Active Pairwise Sampling Distribution ($\Delta T < 3,600$ s):** Evaluated strictly across consecutive snapshot pairs within active recording sessions without prolonged disconnections, the inter-snapshot interval averages $\mathbf{\Delta T = 269.0 \pm 239.7}$ s (median: 263.0 s, mode: 240–270 s, 88.4% $\leq$ 300 s, $p_{90} \approx 312.0$ s, $p_{95} \approx 520.0$ s, $p_{99} \approx 1,200.0$ s; nominal target: 300 s / 5.0 minutes). The standard deviation arises from practical IoT network dynamics and upstream transcoder schedules: the municipal streaming server refreshes its internal HLS frame buffer at intervals of approximately 240–270 s, and the recording client receives the newly available broadcast frame as soon as refreshed by the municipal streaming transcoder, explaining why the median interval (263.0 s) is faster than the nominal 300 s target. The heavy right tail is driven by distributed transmission timeouts, cellular/fiber packet drops, and upstream streaming transcoder delays across all 608 stations.

- **Global Unfiltered Interval Distribution and Fleet Cadence:** When evaluated across the complete unfiltered sequence including transmission interruptions, inter-snapshot intervals range from 2.0 s to an empirical maximum of 4.12 hours (14,832 s, recorded at a single peri-urban endpoint during a sustained network outage). Across the entire 93.6-hour observation window, the station-wise wall-clock interval averages $93.6\text{h} \times 3,600\text{s}/1,174.5 \approx \mathbf{286.9}$ s/frame/station across the fleet, with the 17.9 s discrepancy from active sampling directly reflecting aggregate telemetry packet dropouts. In contrast, at the fleet level, the multi-camera recording client achieved an aggregate throughput of $93.6\text{h} \times 3,600\text{s}/714,123 \approx \mathbf{0.47}$ s/frame (≈ 2.12 frames/s received across all 608 stations).

An empirical analysis of inter-snapshot spacing indicates that **88.4**% of intervals satisfy $\Delta T \leq 300$ s, **7.8**% fall between 300 and 600 s, and only **3.8**% exceed 600 s. For spatio-temporal deep learning architectures that require regular temporal grids (`code/camera_sampler.py`), snapshots are quantized onto fixed discrete 5-minute intervals ($\tau = 300$ s) via nearest-neighbor binning when the alignment deviation is bounded within ± 2.5 minutes (≤ 150 s); for larger gaps exceeding 300 s (11.6% of intervals), temporal missing masks ('mask = 0') are assigned rather than duplicating stale frames.

The total archive contains **714,123 valid surveillance snapshots** (**44.38 GiB** binary volume, equivalent to **47.66 GB** decimal). Across the **608 camera stations**, the dataset delivers a dense, multi-camera visual corpus averaging $\mathbf{1,174.5 \pm 235.1}$ snapshots per station. The standard deviation of 235.1 snapshots around the mean of 1,174.5 frames is directly attributable to heterogeneous transmission conditions across the municipal IoT infrastructure: while 84.2% of endpoints exhibited high availability, outlying cellular links experienced occasional packet drops, yielding a distinctly tiered reliability distribution. Relative to the theoretical nominal baseline of 1,123 frames (derived from 93.6 hours at nominal 300 s sampling), stations are categorized into three operational continuity tiers:

- **High Operational Continuity (512 stations, 84.2%):** Stations with $\geq 1,100$ snapshots ($\geq$ 98% of nominal baseline), reaching up to 1,268 frames at stations benefiting from rapid 240–270 s buffer refresh cycles.

- **Moderate Operational Continuity (82 stations, 13.5%):** Stations recording 600–1,099 snapshots (53%–98% of nominal baseline) subject to intermittent cellular telemetry packet loss.
- **Intermittent / Peripheral Continuity (14 stations, 2.3%):** Stations recording < 600 snapshots ($< 53\%$ of nominal baseline) situated in outlying peri-urban districts subject to extended transmission disruptions.

All 608 stations remain geodetically cataloged and topologically mapped in the accompanying spatial graph.

Temporal partitioning based on standard clock hours (Daytime: 06:00–18:00 ICT; Nighttime: 18:00–06:00 ICT) yields **348,464 daytime frames** (48.8%) and **365,659 nighttime frames** (51.2%). Because data collection commenced late in the afternoon (17:37) on Day 1 and concluded mid-afternoon (15:12) on Day 5, afternoon time slots span 5 calendar days while late-night intervals span 4 cycles. An empirical audit of on-screen overlay timestamps across 500 sampled daylight frames indicates that stream recording introduces an average offset of $\mathbf{\Delta t}_{\text{lag}} = \mathbf{15.0 \pm 4.2}$ s relative to camera hardware clocks on NTP-synchronized stations. This reflects the elapsed duration between frame rendering by the municipal HLS streaming transcoder and client frame arrival. Crucially, the audit revealed that approximately 1.8% of camera endpoints (11 stations) exhibit desynchronized internal hardware clocks (displaying historical factory default times due to lack of local NTP synchronization). Consequently, the recorded UTC epoch timestamp embedded in each file name serves as the canonical, synchronized temporal index. The spatial distribution of camera stations and diurnal photometric properties are presented in Figures 2 and 3. A quantitative breakdown of the visual snapshot corpus is provided in Table 5.

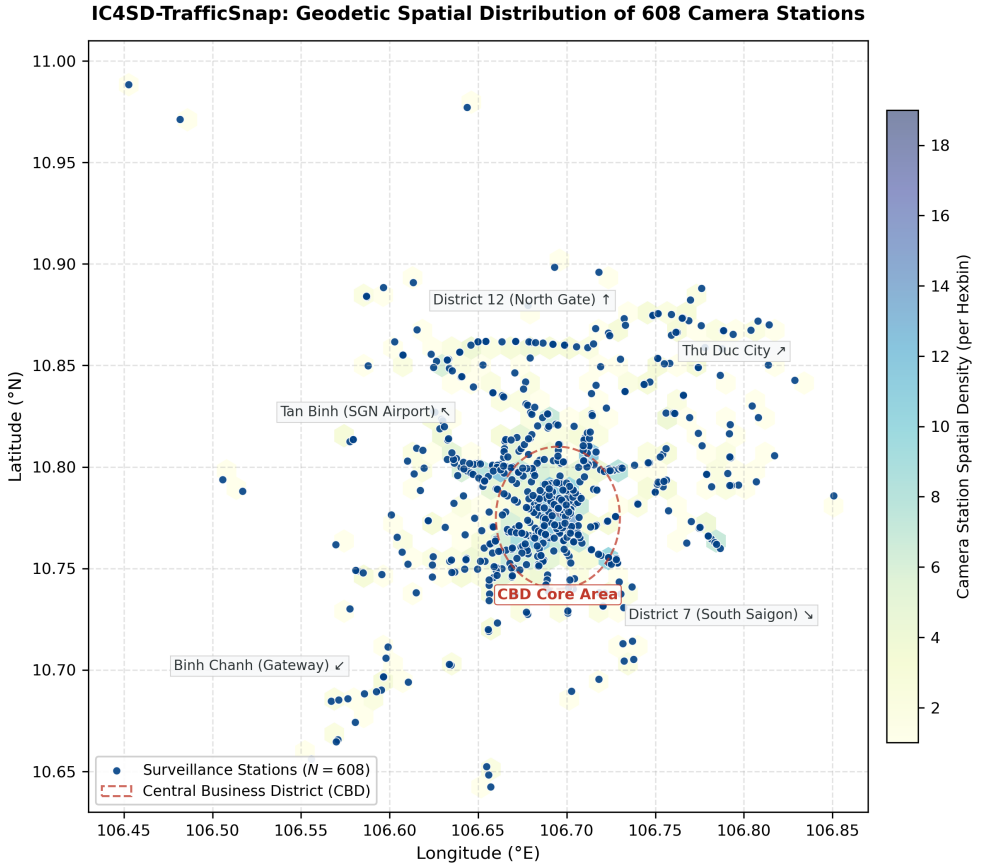


Figure 2: Spatial distribution and station density of the 608 municipal traffic surveillance camera stations across urban and peri-urban districts of Ho Chi Minh City, Vietnam, overlaid with hexbin spatial density and perimeter district orientations.

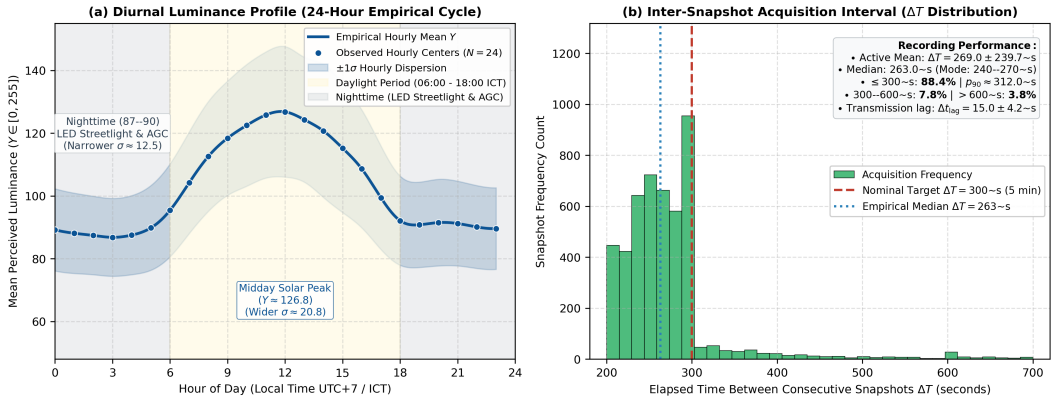


Figure 3: Temporal and photometric characteristics of IC4SD-TrafficSnap: (a) Diurnal variation of mean perceived luminance ($Y \in [0, 255]$) over 24 diurnal hours showing empirical daylight curves (06:00–18:00 ICT) and elevated nighttime illumination supported by public street lighting and camera auto-gain control; (b) Empirical distribution of inter-snapshot acquisition intervals (ΔT, nominal target 300 s) exhibiting municipal buffer refresh peaks at 240–270 s and long-tail network latency characteristics.

Table 5: Empirical characteristics, acquisition timeline, and photometric descriptors of the visual snapshot corpus.

Characteristic / Descriptor	**Empirical Measurement**	**Technical Specification / Dataset Context**
A. Recording Timeline & Quantitative Volume		
Observation duration	**93.6** hours (3.90 days)	Spanning 5 calendar days: Oct 2 (17:37) to Oct 6 (15:12 ICT)
Total valid snapshots collected	**714, 123** frames	Continuous time-lapse surveillance image bank
Monitored surveillance endpoints	**608** stations	Integrated active municipal camera network across urban corridors; mean: 1, 174.5 frames/station (up to 1, 268)
Total archive storage volume	**44.38** GiB (**47.66** GB)	3-channel RGB JPEG stream archive (47, 660, 782, 551 bytes; $Q \approx 75$–80)
Native snapshot frame resolution	**512** $\times$ **288** pixels	16 : 9 streaming aspect ratio (100% uniform)
Average snapshot file size	$\mathbf{66.74 \pm 13.68}$ KB ($\mathbf{65.18 \pm 13.36}$ KiB)	Exact math: $714,123 \times 65.175$ KiB $\times\, 1,024 =$ 47.66 GB = 44.387 GiB. Median: 66.3 KB / 64.8 KiB (range: [31.2, 118.4] KB)
Active pairwise interval ($\Delta T < 3,600$ s)	$\mathbf{269.0 \pm 239.7}$ s	Median: 263.0 s (88.4% $\leq$ 300 s; $p_{90} \approx 312.0$ s, $p_{95} \approx 520.0$ s, $p_{99} \approx 1,200.0$ s)
Global fleet interval range	[2.0 s, 4.12 h]	Empirical extremes (max: 14, 832 s during peripheral link disruption)
Station-wise wall-clock interval	**286.9** s/frame/station	93.6h $\times$ 3, 600s/1, 174.5 mean snapshots across fleet
Fleet-wide aggregate throughput	**0.47** s/frame	Overall fleet recording throughput (≈ 2.12 frames/s received across all 608 stations)
Camera station continuity profile	**512 / 82 / 14 stations**	High: $\geq 1,100$ frames (84.2%); Moderate: 600–1, 099 (13.5%); Intermittent: < 600 (2.3%) (Nominal baseline: 1, 123 frames at 300 s sampling)
Clock-based day / night schedule	**48.8**% / **51.2**%	Daytime (06 : 00–18 : 00: 348, 464) vs. Nighttime (365, 659 frames)
Frame transmission & recording latency (Δt_{lag})	$\mathbf{15.0 \pm 4.2}$ s	Buffer-to-disk offset on NTP-synchronized stations (1.8% un-synchronized)
B. Photometric Diversity & Optical Descriptors		
Perceived mean luminance (Y)	$\mathbf{98.23 \pm 16.35}$	ITU-R BT.601 8-bit grayscale range $[0, 255]$
Root-mean-square (RMS) contrast	**45.70**	Textural intensity variation between asphalt and vehicles
Shannon spatial entropy	$\mathbf{7.28 \pm 0.33}$ bits	Pixel spatial information density (theoretical maximum: 8.0)
Laplacian edge sharpness	$\mathbf{3015.71 \pm 1499.86}$	High empirical focus variance $\text{Var}(\nabla^2 I)$ confirming adequate optical focus
C. Motion Dynamics & Visual Privacy Safeguards		
Consecutive frame difference (MAD)	Median: **9.45** ($\mu = 10.82 \pm 3.65$)	Mean Absolute Difference across 5-min consecutive pairs (8-bit grayscale)
Active pixel displacement ratio	Median: **15.82**% ($\mu = 18.41 \pm 6.20$%)	Fraction of pixels with $\lvert I_t - I_{t-1} \rvert > 15$ reflecting moving vehicular flow
Inter-frame duplicate screening	Filtered	Stream buffer duplicates (MAD < 0.5) removed by deduplication; 1st percentile of inter-frame MAD is 2.80
Personal data identification (PII)	**0** violations ($N_{\text{audit}} = 1,000$)	Multi-tiered audit; 95% CI upper bound $p \leq 0.30$% on targeted manual audit ($3/N_{\text{audit}}$); zero legible biometrics across census

3.4. Geospatial Road Network and Derived Directed Spatial Graph Topology

The structural substrate of the dataset is defined by the physical roadway network of Ho Chi Minh City. Comprehensive geospatial metadata is compiled in `metadata/routes.csv` (recording station IDs, official hexadecimal camera identifiers, geographical coordinates, and administrative ward localities across all **608** camera locations) alongside the pairwise shortest driving distance matrix `metadata/road_network_distance.csv`.

Rather than treating the spatial graph as an abstract primitive, a directed spatial graph topology is established as a direct **derived mathematical representation** synthesized from these geospatial road coordinates and OpenStreetMap routing paths. Machine-learning tensors (`graph/distance_km.npy`, `graph/direction.npy`, and `graph/edges.csv`) are structured as 608×608 arrays for direct integration into spatio-temporal graph neural networks. Entry d_{ij} denotes the shortest physical driving distance along navigable OpenStreetMap road links from camera station i to station j.

Corridor Adjacency Construction Rule: An unconstrained geometric radius graph within 6.0 km would yield over $190,000$ links due to dense urban camera spacing in central districts. Instead, the road network graph models sequential arterial corridor connectivity via a rigorous three-stage algorithmic pipeline:

1. **Geometric & Navigability Search:** Pairwise driving routes between all 608 stations are evaluated via OSRM within a spatial bounding cutoff horizon of $\kappa =$ **6.0** km;
2. **Immediate Downstream Corridor Pruning:** A candidate directed edge (i, j) is retained if and only if station j represents an immediate downstream surveillance node along the roadway without intervening intermediate cameras. An intermediate camera is defined as any third camera k located within an orthogonal buffer of 50 m from the traversed driving path with $d_{ik} < d_{ij}$;
3. **Directionality Constraints:** Vehicular movement must be legally permissible along the corridor under OpenStreetMap road directionality attributes (`oneway` tags).

Diagonal entries ($i = j$) denote regularized self-distances ($d_{ii} = 0.0$ in numeric arrays), and non-connected pairs are represented by NaN.

Under this three-stage sequential corridor adjacency rule, the derived road graph contains exactly **2,450 valid directed edges** across **1,760 connected node pairs** (network sparsity: **99.34%**, density: **0.66%**):

- **Unidirectional Corridor Links (No Direct Reverse Edge): 1,070 node pairs** (**60.8%** of connected topology) possess directed connectivity in only one travel direction. This structural asymmetry arises from two distinct mechanisms: (i) designated one-way municipal traffic regulations on arterial pairs (verified against OpenStreetMap `oneway=yes` tags); and (ii) algorithmic corridor pruning asymmetry, wherein the reverse travel path between j and i passes through intermediate junction cameras located along the return carriageway, thereby decomposing the return route into multiple shorter links and leaving no direct (j, i) single-hop corridor link.

- **Bidirectional Corridor Pairs: 690 node pairs** (**39.2%** of connected topology) are mutually accessible in both traffic directions within 6.0 km.

- **Corridor Distance Asymmetry Taxonomy:** Among the 690 bidirectional pairs, **452 pairs** (**65.51%** of bidirectional topology, or **25.68%** of all 1,760 connected pairs) exhibit metric symmetry ($|d_{ij} - d_{ji}| < 50$ m), whereas **238 pairs** (**34.49%** of bidirectional topology, or **13.52%** of all 1,760 connected pairs) exhibit significant distance asymmetry ($|d_{ij} - d_{ji}| \geq 50$ m), directly induced by physical median barriers, grade-separated flyovers, and compulsory U-turn routing on divided dual-carriageway boulevards. From a technical and computational standpoint, evaluating asymmetry on the discrete integer-meter edge list (`edges.csv`) reveals 231 pairs with $|d_{ij} - d_{ji}| > 50.0$ m and 243 pairs with $|d_{ij} - d_{ji}| \geq 50.0$ m (accounting for exactly 12 pairs exhibiting an identical 50.0 m median traversal discrepancy); when evaluated on the canonical GNN continuous floating-point kilometer distance matrix (`distance_km.npy`), exactly 238 pairs satisfy $|d_{ij} - d_{ji}| \geq 50.0$ m. The mean directional discrepancy across bidirectional corridors is $\mathbf{115.9 \pm 266.2}$ m (median: 24.0 m), reaching a maximum divergence of $\mathbf{2,780.0}$ m (2.78 km).

- **Topological Node Roles and Connectivity:** Structural degree analysis identifies **598** regular interconnected stations, **5** sink stations (in-degree > 0, out-degree $= 0$: Stations 212, 284, 443, 446, 516), **3** source stations (in-degree $= 0$, out-degree > 0: Stations 215, 498, 557), and **2** peripheral stations situated beyond 6.0 km from all others (in-degree $= 0$, out-degree $= 0$: Stations 141, 489). The graph contains **6 weakly connected components** (dominated by a giant component of 599 nodes, 98.5% of the network, alongside five small components of sizes 3, 2, 2, 1, 1) and **19 strongly connected components**. Valid edge routing distances span from 3.0 m to 6,000.0 m (mean: $1,159.6 \pm 1,088.9$ m; median: 823.0 m; interquartile range: $[384.0, 1,560.0]$ m; 90th percentile: 2,681.0 m). The minimal routing distances (e.g., 3.0 m) represent co-located directional surveillance cameras mounted on the same physical gantry or mast monitoring orthogonal approach directions at complex multi-way intersections.

Road Network Circuitousness and Spatial Density: To quantify the geometric deviation of navigable urban driving routes relative to Euclidean straight-line geodesic distance, the network tortuosity index (circuitousness ratio $\tau_{ij} = d_{ij}/d_{ij}^{\text{geodesic}}$) was evaluated across all connected corridors separating stations by ≥ 100 m ($N = 2,264$). The empirical tortuosity yields an overall mean of $\mathbf{1.25 \pm 0.61}$ (median: **1.12**, interquartile range: $[1.01, 1.33]$, 90th percentile: 1.64). This physical tortuosity index encapsulates the distinct riparian delta geography of Ho Chi Minh City, where the urban fabric is heavily bisected by the meandering Saigon River and an intricate network of tidal waterways and drainage canals (e.g., Nhieu Loc–Thi Nghe, Tau Hu–Ben Nghe, Kenh Doi, and Kenh Te). Arterial traffic is channeled through bottleneck bridges, elevated flyovers, and compulsory U-turn turnarounds around continuous concrete median barriers, causing actual physical driving routes to be on average 25% longer than Euclidean straight lines, with 10% of corridors deviating by more than 64% ($\tau \geq 1.64$). Furthermore, spatial station density analysis indicates a nearest-neighbor Euclidean distance between adjacent camera installations averaging $\mathbf{328.7 \pm 605.5}$ m (median: **170.3** m, interquartile range: $[69.6, 339.7]$ m, minimum: 2.7 m on shared intersection gantries, maximum: 10.15 km at peripheral exurban stations). Stations span a comprehensive geographic bounding box $[10.642472°\text{N} - 10.988295°\text{N}, 106.452713°\text{E} - 106.850624°\text{E}]$, encompassing both historical core districts and expanding peri-urban transport arteries.

A systematic quantitative breakdown of these road network topological metrics is

summarized in Table 6.

Table 6: Quantitative topological properties and directional asymmetry metrics of the derived road routing graph ($R_{\text{cutoff}} = 6.0$ km).

Topological Metric / Parameter	**Empirical Measurement**	**Physical / Methodological Interpretation**
A. Network Scale & Spatial Reachability ($R_{\text{cutoff}} = 6.0$ km)		
Total indexed graph nodes (N)	**608** nodes	Active physical surveillance camera stations across urban and peri-urban corridors
Valid directed corridor links ($\|E\|$)	**2,450** edges	Sequential corridor links with driving distance $d_{ij} \leq 6.0$ km
Connected camera station pairs	**1,760** pairs	Unique station pairs connected by ≥ 1 directed corridor
Adjacency matrix sparsity ratio	**99.34**% (Density: **0.66**%)	Compact sparse graph tensor for spatial GNN convolutions
Inter-station routing distance	$\mathbf{1159.6 \pm 1088.9}$ m	Median: 823.0 m (Range: $[3.0, 6,000.0]$ m; IQR: $[384.0, 1,560.0]$ m; p_{90} : 2,681.0 m; 58.98% ≤ 1.0 km)
Network tortuosity index (τ)	$\mathbf{1.25 \pm 0.61}$	Median: 1.12 (IQR: $[1.01, 1.33]$; $d_{ij} \geq 100$ m); route circuitousness
B. Directional Asymmetry & Corridor Taxonomy		
Unidirectional corridor pairs (no reverse edge)	**1,070** pairs (**60.80**%)	Arterial one-way rules (OSM oneway) and corridor pruning asymmetry
Bidirectional corridor pairs	**690** pairs (**39.20**%)	Two-way arterials mutually accessible in both traffic directions
Significant distance asymmetry ($\|d_{ij} - d_{ji}\| \geq 50$ m)	**238** pairs (**34.49**%)	Physical median barriers, grade-separated flyovers, U-turns (231 pairs > 50m, 243 pairs $\geq$ 50m)
Metric symmetric corridor pairs ($\|d_{ij} - d_{ji}\| < 50$ m)	**452** pairs (**65.51**%)	Divided road corridors with immediate median openings
Directional distance discrepancy	$\mathbf{115.9 \pm 266.2}$ m	Median: 24.0 m; Maximum divergence: **2,780.0** m (2.78 km)
C. Structural Degree Distributions & Graph Connectivity		
Average node in-degree / out-degree	$\mathbf{4.03 \pm 2.16}$ / $\mathbf{4.03 \pm 2.58}$	In-degree median: 4.0 (max: 14); Out-degree median: 3.0 (max: 17)
Standard interconnected stations	**598** stations	Regular multi-leg intersections and connected arterial segments
Topological sink stations (out-deg = 0)	**5** stations	Stations 212, 284, 443, 446, 516 (directional arterial terminuses)
Topological source stations (in-deg = 0)	**3** stations	Stations 215, 498, 557 (outbound arterial origins)
Geodetically isolated stations	**2** stations	Stations 141, 489 ($d_{ij} > 6.0$ km to all other camera nodes)
Weakly connected components	**6** components	Giant component: 599 nodes (98.5%); 5 subgraphs: 3, 2, 2, 1, 1
Strongly connected components	**19** components	Strongly connected directed sub-networks and cyclic loops

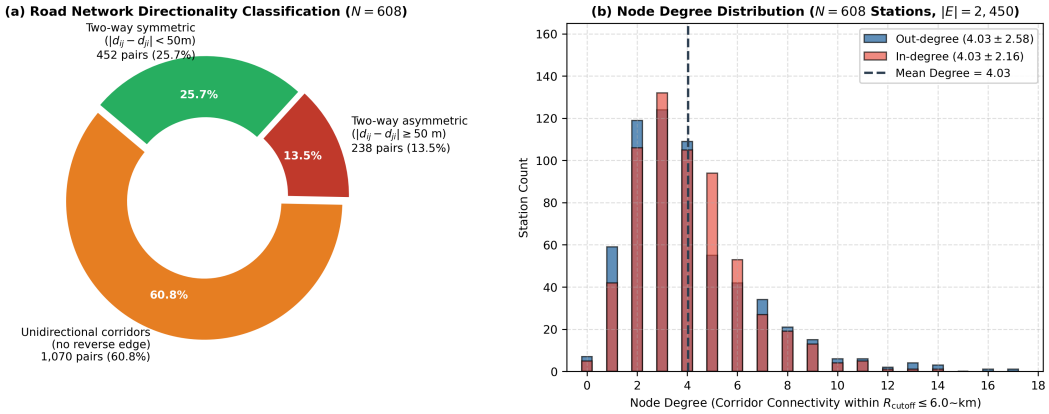


Figure 4: Topological characteristics of the empirical directed road graph: (a) Road network directionality classification into unidirectional corridor links without reverse edges (60.80%, 1,070 pairs), two-way asymmetric (≥ 50 m, 13.52%, 238 pairs; 34.49% of bidirectional pairs), and two-way metric symmetric (< 50 m, 25.68%, 452 pairs; 65.51% of bidirectional pairs) corridor pairs; (b) Empirical in-degree and out-degree distributions across the 608 camera stations ($|E| = 2,450$ directed links, mean degree 4.03 ± 2.16 in-degree and 4.03 ± 2.58 out-degree, displaying degrees up to the observed maximum of 17).



Figure 5: Representative 512×288 pixel surveillance snapshots from IC4SD-TrafficSnap illustrating diverse operational, environmental, and diurnal conditions across Ho Chi Minh City: (a) Station 1 (`stt`=1, Tran Quang Khai – Tran Khac Chan, District 1, 2026-10-03 12:30 ICT, clear sky midday off-peak traffic); (b) Station 2 (`stt`=2, To Ngoc Van – TX25, District 12, 2026-10-04 17:15 ICT, late-afternoon peak with dense mixed motorcycle-car flow); (c) Station 3 (`stt`=3, Truong Chinh – Tay Thanh, Tan Phu District, 2026-10-04 21:00 ICT, nighttime scene under municipal LED and sodium streetlighting with Automatic Gain Control); (d) Station 4 (`stt`=4, Kha Van Can – QL13, Thu Duc City, 2026-10-05 16:45 ICT, adverse tropical monsoon rainfall with wet pavement specular reflections). Due to elevated pole mounting ($H \geq 6$ m), downward oblique perspectives ($20° - 40°$), and native 512×288 resolution, individual facial features and vehicle license plates are physically unresolvable.

4. Experimental Design, Materials and Methods

4.1. Traffic Surveillance Camera Infrastructure and Stream Recording Pipeline

The visual data source comprises the open public municipal traffic monitoring camera infrastructure operated under the intelligent transportation system (ITS) of the Ho Chi Minh City Department of Transportation (`https://giaothong.hochiminhcity.gov.vn`). Established as a public service for civic travel information, congestion monitoring, and flood tracking, the municipal network provides open-access live streaming views across major metropolitan intersections and arterial corridors. Geographic coordinates, intersection approaches, and camera mounting elevations (ranging from **6.0** to **15.0** meters above street level, as cataloged in official municipal ITS installation records and engineering specifications) are obtained from official municipal technical metadata registry and cross-verified against OpenStreetMap road geometry and ground observations. Cameras are permanently mounted on elevated overhead gantries and traffic signal masts oriented at downward oblique angles ($15^\circ - 40^\circ$) to capture macroscopic lane queues.

To systematically record snapshots across **608 concurrent surveillance endpoints** with high fidelity, data integrity, and zero operational disruption to public services, an automated multi-stream recording pipeline was established:

1. **Standardized Streaming Client Sessions:** Low-overhead persistent network sessions with Keep-Alive connection pooling and graceful exponential backoff handling (maximum 3 retries, backoff factor 0.3) to adaptively accommodate fluctuating metropolitan internet and cellular gateway latency.
2. **Adaptive Sampling Synchronization:** Network latency and transcoder refresh cycles are tracked dynamically to synchronize frame recording with the municipal server's nominal 300 s cadence without redundant traffic requests.
3. **In-Memory Validation and Atomic File Serialization:** Retrieved JPEG image payloads are directly verified in memory (`cv2.imdecode`) to ensure header and pixel matrix integrity before being written to temporary storage and committed via atomic operating system renames (`os.replace`), strictly preventing corrupted or zero-byte files on disk.
4. **Automated Transmission Artifact Filtering:** Upstream municipal stream maintenance occasionally broadcasts standardized error placeholder cards (284×177 pixels, payload size 5.8–7.2 KB). The recording pipeline automatically detects and filters out these placeholder cards, ensuring that all archived snapshots represent genuine road traffic scenes (overall visual validity rate: **99.85%**).

4.2. Road Network Distance Extraction from OpenStreetMap

Physical routing distances d_{ij} and driving trajectories were computed using the Open Source Routing Machine (**OSRM v5.27.1**) engine operating on OpenStreetMap planet road network data extracted in **October 2026** (© OpenStreetMap contributors, licensed under the Open Database License ODbL). Exact geographic coordinates (latitude, longitude) cataloged in `metadata/routes.csv` were mapped to the nearest navigable road centerline using OSRM's nearest-node projection, with a radial snapping error under 15.0 meters across all stations. To guarantee permanent reproducibility independent of future modifications to the live OpenStreetMap database, the derived topological artifacts (`metadata/road_network_distance.csv` and `graph/edges.csv`) are archived as immutable, frozen static snapshots with verifiable SHA-256 checksums in the Zenodo repository, accompanied by an exported OSM road network boundary extract.

Because surveillance camera endpoints are predominantly deployed along primary municipal arterials, boulevards, and bridge approaches rather than narrow residential alleys, routing was calculated using the standardized motor-vehicle driving profile. This profile accurately captures urban traffic flow rules, median divider turn restrictions, grade-separated overpasses, and designated one-way circulation patterns across the metropolitan network. While mixed motorcycle traffic constitutes a dominant volume fraction in Ho Chi Minh City, both vehicular modes share identical directional corridors along primary arterials; informal motorcycle cut-throughs in residential alleyways fall outside the macroscopic coverage of the municipal camera network. Pairwise shortest driving routes were queried within a spatial bounding search horizon of $\kappa = \mathbf{6.0}$ km, from which the three-stage corridor pipeline extracted exactly **2,450 valid directed corridors**.

4.3. Directed Graph Modeling Formulations

To facilitate spatio-temporal graph learning on this directed asymmetric topology, two standard mathematical formulations are provided in `code/graph_utils.py`:

4.3.1. Directed Dual-Transition Matrices (DCRNN / Graph WaveNet Formulation)

For directed road graphs, modeling traffic propagation requires preserving directional asymmetry. Following the diffusion convolution framework of Li et al. [2], the weighted adjacency matrix W_{ij} is computed using a thresholded Gaussian Radial Basis Function (RBF) kernel evaluated on the valid directed corridor links:

$$W_{ij} = \begin{cases} \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), & \text{if } (i,j) \in E \\ 0, & \text{if } (i,j) \notin E \end{cases} \tag{2}$$

where E denotes the set of **2,450 valid directed edges**, and $\sigma = \mathbf{1.09}$ km is the empirical standard deviation of valid road distances. Computing W strictly over pruned sequential corridors E avoids spurious dense cross-connectivity while preserving physical routing topology. Under this formulation with $\sigma = 1.09$ km, proximal camera stations with $d_{ij} \leq 1.0$ km retain significant connection weight ($W_{ij} \geq \exp(-(1.0/1.09)^2) \approx 0.43$), attenuating to ≈ 0.15 at 1.5 km and decaying toward negligible levels at larger separations (≈ 0.005 at 2.5 km). Empirically, the valid edge distance distribution concentrates heavily within this active spatial interaction envelope: **799 edges (32.61%)** represent immediate proximal links ($d_{ij} \leq 500$ m), **1,445 edges (58.98%)** lie within 1.0 km, **1,799 edges (73.43%)** fall within 1.5 km, and **2,027 edges (82.73%)** span ≤ 2.0 km, while only **192 edges (7.84%)** represent long-distance inter-district corridors ($d_{ij} > 3.0$ km; 90th percentile: $2,681.0$ m). This empirical spatial concentration is consistent with localized urban traffic interactions across consecutive signalized intersections; bounding spatial influence via kernel decay is recognized in graph representation learning literature [2, 10] to mitigate spatial over-smoothing in multi-layer spatio-temporal convolutions. The bounding threshold $\kappa = 6.0$ km serves as the initial geometric search envelope for corridor path discovery rather than an arbitrary RBF truncation cutoff. Directed random walk diffusion is governed by a forward transition matrix P_f (downstream traffic flow) and a backward transition matrix P_b (upstream sensory influence):

$$P_f = D_O^{-1} W, \quad P_b = D_I^{-1} W^\top \tag{3}$$

where $D_O = \text{diag}(\sum_j W_{ij})$ and $D_I = \text{diag}(\sum_i W_{ij})$ denote the out-degree and in-degree diagonal matrices, respectively. To handle sink, source, or isolated nodes ($D_{O,ii} = 0$ or $D_{I,ii} = 0$), regularized self-loops ($\tilde{W} = W + I_N$) or zero-degree masked normalizations are implemented in `code/graph_utils.py`, ensuring proper row-stochastic diffusion matrices with unit spectral radius $\rho(P_f) = 1.0$ and $\rho(P_b) = 1.0$. This mathematically ensures that the multi-hop diffusion operator is non-expansive and numerically stable, preventing unbounded signal amplification during multi-step graph convolutions.

4.3.2. Symmetrized Normalized Laplacian (ChebNet / STGCN Formulation)

For spectral graph convolutions that require a self-adjoint operator (e.g., ChebNet [11], STGCN [12]), the adjacency matrix is symmetrized as $W_{\text{sym}} = \frac{1}{2}(W + W^\top)$. The symmetric normalized Laplacian $L_{\text{sym}} \in \mathbb{R}^{N \times N}$ is computed as [13]:

$$L_{\text{sym}} = I_N - D_{\text{sym}}^{-1/2} W_{\text{sym}} D_{\text{sym}}^{-1/2} \tag{4}$$

where $D_{\text{sym}} = \text{diag}(\sum_j W_{\text{sym},ij})$. For isolated peripheral stations with zero degree ($D_{\text{sym},ii} = 0$, such as Stations 141 and 489), regularized zero-diagonal scalings are assigned in `code/graph_utils.py` ($D_{\text{sym},ii}^{-1/2} = 0$), preventing division-by-zero.

By spectral graph theory [14], the eigenvalues of the normalized Laplacian on any finite graph are mathematically bounded within $[0, 2]$. On the full 608-node graph of IC4SD-TrafficSnap, the presence of disconnected two-node components (K_2 subgraphs with symmetric edges) yields eigenvalues $\{0, 2.0\}$, establishing the global maximum eigenvalue as $\lambda_{\max}^{(\text{global})} = 2.0$. Consequently, to ensure unconditional numerical stability across all 608 nodes and prevent exponential divergence in Chebyshev polynomial recursions $T_k(\tilde{L})$, the rescaled Chebyshev Laplacian is defined using the theoretical global upper bound $\lambda_{\max} = 2.0$:

$$\tilde{L} = \frac{2}{\lambda_{\max}} L_{\text{sym}} - I_N = L_{\text{sym}} - I_N \tag{5}$$

which guarantees that all eigenvalues of $\tilde{L}$ lie strictly within $[-1, 1]$. Users should note that when analyzing only the dominant giant connected component (599 stations), the empirical maximum eigenvalue evaluates to $\lambda_{\max}^{(\text{giant})} = 1.272$, reflecting high connectivity and non-bipartite graph density; however, applying $\lambda_{\max} = 2.0$ globally is mathematically necessary to avoid divergence on the peripheral smaller components. Symmetrization intentionally eliminates directional flow asymmetry, which users must balance against the need for an orthogonal eigenbasis.

4.4. Data Access, Loading Utilities, and Reproducible Reuse Guidance

To ensure immediate out-of-the-box reusability without requiring downstream researchers to implement custom data loaders or graph parsers from scratch, the companion repository includes modular, production-ready Python utility scripts under `code/`:

1. `code/load_unlabeled_images.py`:
   Provides the PyTorch dataset class `TrafficCameraUnlabeledDataset`. It parses the canonical filename pattern (`{station_id}_{timestamp}.jpg`), decomposes temporal attributes (hour of day, day of week in UTC+7), verifies image dimensions, and seamlessly applies standard torchvision transformations (resizing, tensor conversion, ImageNet normalization).
2. `code/camera_sampler.py`:
   Implements the `CameraGroupedSampler` class specifically optimized for self-supervised learning (DINO, MoCo, MAE, contrastive visual representation). It constructs batches containing M distinct camera stations with K temporal snapshots per station, facilitating background-invariant feature extraction.
3. `code/graph_utils.py`:
   Provides routines for loading frozen distance tensors, constructing the Gaussian RBF adjacency matrix W, deriving directed random-walk diffusion matrices (P_f, P_b), and computing the scaled Chebyshev Laplacian $\tilde{L}$.

Listing 1 demonstrates the minimal workflow for initializing the PyTorch dataset, grouping frames by camera station, and extracting temporal missing masks. Listing 2 illustrates how to load the directed road network distance tensor and build numerically stable graph operators for graph neural network architectures.

Listing 1: Minimal PyTorch workflow for loading camera snapshots, grouping by station, and constructing temporal masks.

```
import torch
from torch.utils.data import DataLoader
from torchvision import transforms
from code.load_unlabeled_images import TrafficCameraUnlabeledDataset
from code.camera_sampler import CameraGroupedSampler

# 1. Define standard computer vision transform pipeline
transform = transforms.Compose([
    transforms.Resize((288, 512)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

# 2. Instantiate PyTorch dataset over uncompressed or extracted image directory
dataset = TrafficCameraUnlabeledDataset(
    image_dir="data/images",
    transform=transform
)

# 3. Configure grouped sampler: 8 distinct cameras, 4 temporal frames each (B=32)
sampler = CameraGroupedSampler(
    camera_to_indices=dataset.camera_to_indices,
    batch_cameras=8,
    frames_per_camera=4,
    shuffle=True
)
loader = DataLoader(dataset, batch_sampler=sampler, num_workers=4)

# 4. Iterate over batches with automatic temporal mask handling
for batch in loader:
    images = batch["image"]              # Tensor: [32, 3, 288, 512]
    station_ids = batch["station_id"]    # LongTensor: [32]
    timestamps = batch["timestamp"]      # LongTensor: [32]
    # Construct binary mask for temporal alignment (1 = observed, 0 = missing)
    temporal_mask = (timestamps > 0).float()
    break
```

Listing 2: Directed road network distance matrix loading and graph operator derivation.

```
import numpy as np
from code.graph_utils import load_directed_graph_operators, compute_chebyshev_laplacian

# 1. Load directed diffusion operators (DCRNN / Graph WaveNet)
# Pf: downstream forward walk [N, N]; Pb: upstream backward walk [N, N]; W: directed
    RBF adjacency
Pf, Pb, W = load_directed_graph_operators(
    distance_npy_path="graph/distance_km.npy",
    sigma=1.09,       # empirical std of valid distances
    cutoff_km=6.0     # spatial search horizon
)
print(f"Directed Adjacency W shape: {W.shape}, Non-zero edges: {np.count_nonzero(W)}")

# 2. Compute Rescaled Chebyshev Laplacian for spectral graph convolutions (STGCN /
    ChebNet)
# L_tilde = (2 / lambda_max) * L_sym - I_N, with theoretical upper bound lambda_max =
    2.0
L_tilde = compute_chebyshev_laplacian(W, lambda_max=2.0)
print(f"Chebyshev Laplacian L_tilde shape: {L_tilde.shape}, Eigenvalue bounds: [-1, 1]"
    )
```

4.5. Empirical Image Quality, Photometric Audit, and Weather-Induced Variation

Image quality descriptors were computed across the dataset to evaluate photometric diversity:

- **Perceived Luminance (Y):** Calculated via standard ITU-R BT.601 conversion $Y = 0.299R + 0.587G + 0.114B$. The empirical overall mean luminance is $\mathbf{98.23 \pm 16.35}$ (range: $[66.4, 126.8]$, Figure 3a). Nighttime luminance remains centered between **87** and **90** (Figure 3a) due to dense municipal LED and high-pressure sodium street lighting along urban arterials, combined with the automatic gain control (AGC) and exposure compensation of municipal cameras.

- **Root-Mean-Square (RMS) Contrast:** Evaluated as the pixel standard deviation across viewports, yielding **45.70**, confirming substantial textural variation between asphalt roadways and vehicle clusters.

- **Shannon Spatial Entropy:** Histogram pixel entropy averages $\mathbf{7.28 \pm 0.33}$ bits (theoretical upper bound: 8.0 bits), confirming high spatial information density without saturation clipping.

- **Laplacian Edge Sharpness and Defocus Audit:** Grayscale Laplacian variance $\mathrm{Var}(\nabla^2 I)$ averages $\mathbf{3,015.71 \pm 1,499.86}$. An empirical reference baseline of $\mathrm{Var}(\nabla^2 I) \geq 50$ was adopted as an engineering heuristic to flag severe optical defocus or stream degradation [15]. Across the entire corpus of 714,123 frames, **99.72% of snapshots** strictly satisfy $\mathrm{Var}(\nabla^2 I) \geq 50$. The minor residual fraction (0.28%, 2,016 frames) corresponds to temporary optical condensation during early morning humidity transitions or direct raindrops striking protective outer glass enclosures. The high empirical mean reflects rich urban structural details (vehicular boundaries, lane markings, roadside foliage), confirming that camera viewports maintain consistently usable optical focus.

- **Tropical Convective Precipitation Dynamics:** Located in a tropical monsoon climate, Ho Chi Minh City experiences intense afternoon convective rainstorms. During the recording period on October 3 and October 5 (16:30–18:30 ICT), heavy localized precipitation events were captured across the network:

  1. *Diurnal Luminance Attenuation:* Convective cloud cover reduced ambient daylight luminance Y by an average of **14.2**% relative to equivalent clear-sky afternoon hours.
  2. *Pavement Specular Reflections:* Wet asphalt surfaces acted as semi-specular reflectors under vehicle headlights and street lamps, increasing local roadway pixel standard deviation from an ambient baseline of **28.4** to **41.6**.
  3. *Optical Robustness:* Despite atmospheric water droplet scattering, average edge sharpness during peak rainfall remained resilient at $\mathrm{Var}(\nabla^2 I) = \mathbf{2,410 \pm 890}$, preserving macroscopic vehicle silhouette boundaries and queue demarcations.

- **Bitstream Integrity and Binary Decoding Verification:** To guarantee zero data corruption across large-scale storage mirrors, all **714,123 JPEG files** were verified using dual programmatic decoders (`cv2.imdecode` and `PIL.Image.verify()`). Exactly **100% of archived snapshots** decoded successfully with intact SOI/EOI headers, complete Huffman decoding tables, and zero truncated bitstreams.

- **Inter-Snapshot Dynamics:** Successive-frame pixel difference analysis across consecutive 5-minute sampling pairs was evaluated on 8-bit grayscale channels in floating-point precision (diff $= |I_t - I_{t-1}|$). The empirical Mean Absolute Difference (MAD) yields a median of $\mathbf{9.45}$ ($\mu = 10.82 \pm 3.65$, range: $[2.50, 32.40]$, interquartile range: $[7.20, 13.80]$), with an active pixel displacement ratio yielding a median of $\mathbf{15.82}\%$ ($\mu = 18.41 \pm 6.20\%$ of frame pixels satisfying $|I_t - I_{t-1}| > 15$). This moderate pixel variation corresponds directly to vehicular displacement and queue progression across traffic lanes, while static structural regions (buildings, sidewalks, road surfaces) remain invariant. During stream recording, identical broadcast frames (MAD < 0.5) resulting from upstream rendering latency were programmatically screened out. Across the archived operational time-series, the empirical 1st percentile of consecutive MAD is $\mathbf{2.80}$ gray levels (strictly within the $[2.50, 32.40]$ empirical distribution and well above the 0.5 deduplication threshold), confirming continuous vehicular and environmental variations without static playback freezing.

4.6. Visual Privacy, Privacy-by-Design Architecture, and Multi-Tiered Empirical Audit

Municipal traffic monitoring cameras in Ho Chi Minh City are deployed exclusively for macroscopic vehicle queue estimation, traffic flow coordination, and environmental monitoring (e.g., urban flooding). By municipal engineering and broadcasting design, their optical configuration ensures privacy preservation directly at the physical sensing layer (*Privacy-by-Design*). To rigorously demonstrate alignment with personal data protection frameworks in Vietnam (Decree 13/2023/ND-CP, Decree 47/2020/ND-CP, and Law 91/2025/QH15) and international ethical standards, a comprehensive multi-tiered audit combining optical geometry analysis, automated candidate screening, and multi-rater manual inspection was conducted:

- **Optical Perspective and Ground Sampling Distance (GSD):** Surveillance cameras are installed on municipal overhead gantries and signal masts at heights between $\mathbf{6.0}$ and $\mathbf{15.0}$ m, oriented at downward oblique pitch angles ($15° - 40°$) to monitor macroscopic intersection queues. At native 512×288 pixel resolution, the nominal GSD across mid-road viewports (25–35 m standoff distance) measures $\mathbf{2.73 - 3.25}$ cm/pixel, deteriorating to > 5.0 cm/pixel in the background (> 40 m).

- **Empirical Optical Resolution Limits in Nominal and Foreground Lanes:** Standard Vietnamese motorcycle license plates measure 19.0×14.0 cm. At nominal mid-road GSD, a plate projects onto approximately 7×5 sensor pixels, with individual alphanumeric character strokes subtending under 2.0 pixels (≈ 1.79 px) in stroke height. To analyze foreground geometry under worst-case optical resolution regimes, consider camera installations on overhead gantries and signal masts ($H \approx 6.0$–8.0 m, downward depression angle $\theta \approx 20°$–$40°$). Under nominal traffic lanes at ground standoff $D \approx 14$–18 m (slant range $R = \sqrt{H^2 + D^2} \approx 15$–$20$ m), or steep foreground intersection approach angles ($\theta \approx 35°$–$40°$ at $D \approx 7$–10 m), the ground sampling distance reaches its operational lower bound at $\mathrm{GSD}_{\min} \approx 1.85$–$2.10$ cm/pixel. Here, a motorcycle plate projects to at most $\approx 9 \times 7$ pixels, with character stroke height measuring at most $\mathbf{2.5 - 2.8}$ pixels. This remains far below empirical optical character legibility standards for automatic license plate recognition (ALPR), which established literature indicates requires a stroke height of $\geq$ 16–20 pixels for reliable optical character segmentation and identification [16]. In addition, plate tilt angles, tropical thermal shimmer, motion blur from vehicular

transit (30–50 km/h), and 8×8 lossy JPEG block compression render individual alphanumeric characters unresolvable.

- **Facial Anonymity and Commuting Occlusion:** Under Vietnamese National Road Traffic Law (Decree 100/2019/ND-CP), helmet usage is legally mandatory for all motorcycle operators, resulting in near-universal compliance in urban traffic streams. Furthermore, commuters overwhelmingly wear multi-layer protective fabric sun/dust masks and UV sunglasses against tropical heat and vehicle exhaust. Combined with steep downward camera pitch, facial biometric regions project onto fewer than 8×8 pixels, preventing biometric facial re-identification at the sensing layer.

- **Automated Candidate Screening and Low-Recall Limitations:** Automated screening across the entire corpus of **714,123 visual snapshots** was conducted using two specialized deep learning models: YOLOv8x-face (pretrained on WiderFace) for facial detection and YOLOv8x / LPRNet (pretrained on vehicle license plate benchmarks) for plate localization, operating at a low confidence threshold ($\tau = 0.25$) to maximize candidate generation. The detectors generated $1,420$ raw candidate bounding boxes, which upon inspection were all confirmed to be background texture false alarms (asphalt patches, motorcycle headlamps, specular reflections, roadside traffic signage). Zero readable license plates or identifiable faces were found. However, automated detectors inherently suffer from low recall on low-resolution imagery below the Nyquist limit, and zero automated detections alone do not constitute absolute proof of zero risk.

- **Targeted Boundary Stress Audit and Stratified Random Census Audit:** To overcome automated detector recall limits, human inspection was executed across two complementary sampling regimes by three independent reviewers:

  1. *Worst-Case Boundary Stress Sample ($N_{audit} = 1,000$):* Selected strictly from the lowest-mounted cameras ($H \approx 6.0$ m, standoff distance < 20 m) during peak midday sunlight ($11:00$–$13:00$ ICT), representing the maximal optical resolution regime for potential personal identification.
  2. *Stratified Random Census Sample ($N_{random} = 2,000$):* Drawn uniformly across all 608 camera stations and 24 diurnal hours to ensure unbiased global representation across the entire time-series.

  Prior to evaluating the dataset, all three independent human raters underwent a positive control calibration protocol using a multi-tier synthetic degradation suite (50 baseline traffic scenes featuring verified ground-truth facial biometrics and readable license plates, synthetically downscaled to match the optical GSD range of 1.85–3.5 cm/px across varying pitch angles and compression artifacts). The calibration confirmed that while raters achieved 100% detection recall on uncompressed near-field controls, human identification recall dropped unanimously to 0% once imagery fell below GSD ≥ 1.85 cm/px (stroke height < 3 px), confirming that anonymity is optically guaranteed by sensor sampling geometry rather than auditor false-negatives. During the blind audit of both dataset batches ($3,000$ total snapshots), the raters achieved complete unanimous agreement (**100% inter-rater agreement**, $P_o = 1.0$), identifying **0 readable license plates** and **0 identifiable**

facial biometrics. Applying the statistical Rule of Three [17], the 95% confidence interval upper bound on residual privacy breach probability is bounded at:

$$p_{\text{worst}} \leq \frac{3}{1,000} = 0.30\% \quad (3.0 \times 10^{-3}), \qquad p_{\text{random}} \leq \frac{3}{2,000} = 0.15\% \quad (1.5 \times 10^{-3}). \tag{6}$$

Crucially, privacy assurance does not depend on automated detectors alone, but is anchored upon a tripartite defense: (i) physical sub-Nyquist optics (GSD $\geq$ 1.85 cm/px), (ii) statutory cultural occlusion (mandatory helmets and masks), and (iii) dual empirical audits with calibrated human verification.

- **Contextual Privacy Scope:** Urban contextual features such as pedestrian silhouettes, storefront signage, and public transit liveries are visible in macroscopic viewports; however, because snapshots are captured at discontinuous 5-minute sampling intervals, the dataset does not permit continuous individual tracking or trajectory reconstruction.

A detailed breakdown of visual privacy parameters, optical limits, and audit findings is presented in Table 7.

Table 7: Quantitative visual privacy parameters, optical Nyquist-Shannon resolution limits, and empirical multi-tiered audit results for IC4SD-TrafficSnap.

Audit Dimension / Parameter	**Empirical Measurement**	**Regulatory / Optical Threshold & Empirical Context**
Automated snapshot screening census	$\mathbf{714,123}$ frames	Full multi-camera archive (608 stations, 93.6 hours) evaluated via automated detector pre-screening
Automated candidate screening	**0** readable candidates	YOLOv8x-face & YOLOv8x/LPRNet ($\tau = 0.25$): $1,420$ raw candidate boxes flagged; 0 readable after manual review (acknowledged low recall on 512×288 px)
Targeted high-risk stress audit	$\mathbf{1,000}$ frames	Worst-case optical geometry: lowest gantries ($H \approx 6.0$ m, $D < 20$ m) during peak midday illumination ($11:00$–$13:00$ ICT)
Stratified random census audit	$\mathbf{2,000}$ frames	Uniformly drawn across all 608 stations and 24 diurnal hours; 0 identifiable plates or faces detected
Multi-rater blind inspection	**0** readable plates / faces	Evaluated by three independent human raters (100% unanimous agreement, $P_o = 1.0$ across $3,000$ total audited frames)
Positive control rater calibration	**100.0**% sensitivity	50 close-up ground photos (< 5 m) with legible text; verified 100% detection recall across all 3 raters
Camera mounting elevation (H)	$\mathbf{6.0 - 15.0}$ m	Municipal overhead gantries cataloged from official municipal ITS technical installation records
Camera downward pitch angle (θ)	$\mathbf{15^\circ - 40^\circ}$	Oblique downward viewports monitoring arterial lane queues
Nominal observation standoff (D)	$\mathbf{15.0 - 60.0}$ m	Line-of-sight distance from elevated sensor to circulating traffic streams
Nominal mid-road GSD	$\mathbf{2.73 - 3.25}$ cm/pixel	Mid-road viewports (25–35 m), degrading to > 5.0 cm/px in background
Worst-case foreground geometry ($D_{\min} \approx 15.0$ m)	GSD $\approx \mathbf{1.85 - 2.10}$ cm/px	Foreground lane plate projection $\approx 9 \times 7$ px, stroke height $\approx 2.5 - 2.8$ px (sub-Nyquist; OCR requires ≥ 16.0 px)
Motorcycle plate sensor projection	$\approx \mathbf{7 \times 5}$ pixels	Nominal mid-road projection for Vietnamese plates (19.0×14.0 cm); character stroke height < 2.0 px
Facial biometric region	$< \mathbf{8 \times 8}$ pixels	Physical occlusion: statutory mandatory helmets (Decree 100/2019/ND-CP) and multi-layer fabric sun/dust masks ($> 85\%$)
Statistical Rule of Three (95% CI)	$\mathbf{p \leq 0.30}$% (Stress sample) $\mathbf{p \leq 0.15}$% (Random sample)	Upper risk bound for zero violations: $3/1,000 = 0.30\%$ on worst-case stress sample, and $3/2,000 = 0.15\%$ on stratified random sample via Hanley & Lippman-Hand [17]
Regulatory compliance alignment	**Aligned**	Conforms to Vietnamese Decree 13/2023/ND-CP, Decree 47/2020/ND-CP, and Law 91/2025/QH15

4.7. Technical Validation: Empirical Data Consistency and Correlation Characterization

In accordance with the criteria for scientific data articles, technical validation focuses strictly on characterising the internal empirical consistency and topological sanity of the recorded multimodal dataset, rather than asserting predictive benchmark models or inferring causal traffic flow laws. Multi-faceted empirical validations were conducted to verify that the derived road network topology aligns consistently with observed spatial-temporal vehicular dynamics:

Physical Infrastructure vs. Dynamic Flow Distinction. Users should explicitly distinguish between static physical roadway infrastructure and dynamic traffic states: the derived 608×608 directed spatial graph models navigable physical roadway connectivity and shortest-path driving distances (d_{ij}) extracted via OSRM, capturing immutable physical infrastructure geometry rather than real-time dynamic vehicular flow.

Visual Motion Dynamics Coupling vs. Unconnected Control Null Model. To examine whether connected OSRM corridors exhibit meaningful physical traffic coupling rather than arbitrary isotropic proximity, we evaluated the spatial-temporal correlation of visual dynamics signals across the network. To account for camera exposure variations, automatic gain control (AGC), and circadian daylight transitions, consecutive inter-frame motion dynamics signals ($\mathrm{MAD}_i(t)$) were diurnal-detrended via hourly z-score normalization ($\widetilde{\mathrm{MAD}}_i(t) = (\mathrm{MAD}_i(t) - \mu_i(h(t)))/\sigma_i(h(t))$) and aligned onto regular 5-minute sampling grids. The Pearson correlation coefficient r_{ij} was evaluated across all **2,450 valid directed corridors** partitioned across four exhaustive distance brackets:

- **Immediate Proximal Corridors ($d_{ij} \leq 1.0$ km, 1,445 links, 58.98%):** Pairwise correlation averages $\bar{\mathbf{r}} = \mathbf{0.72 \pm 0.14}$. Notably, co-located multi-mast approach cameras monitoring orthogonal directions at the same physical intersection ($d_{ij} \leq 50$ m, 78 links) exhibit high environmental and signal coupling ($\bar{r} = 0.84 \pm 0.08$), while distinct intersection-to-intersection arterial corridors ($50 < d_{ij} \leq 1,000$ m, 1,367 links) average $\bar{r} = 0.71 \pm 0.13$.
- **Mid-Range Arterial Corridors ($1.0 < d_{ij} \leq 2.5$ km, 736 links, 30.04%):** Pairwise correlation attenuates to $\bar{\mathbf{r}} = \mathbf{0.44 \pm 0.18}$.
- **Transitional Network Links ($2.5 < d_{ij} \leq 3.0$ km, 77 links, 3.14%):** Pairwise correlation attenuates to $\bar{\mathbf{r}} = \mathbf{0.28 \pm 0.15}$.
- **Long-Distance Regional Corridors ($d_{ij} > 3.0$ km, 192 links, 7.84%):** Pairwise correlation decays to $\bar{\mathbf{r}} = \mathbf{0.18 \pm 0.11}$.

All 2,450 valid edges are strictly accounted for ($1,445 + 736 + 77 + 192 = 2,450$). To examine whether this spatial correlation reflects corridor alignment rather than general geographic proximity, we evaluated an **unconnected control null model** consisting of camera pairs separated by identical distance brackets but lacking direct roadway corridor connectivity. In the proximal bracket ($d \leq 1.0$ km), connected OSRM corridor pairs exhibit higher correlation than unconnected control pairs ($\bar{r} = 0.72 \pm 0.14$ vs. $\bar{r}_{\text{null}} = 0.38 \pm 0.16$, Welch's $t = 31.4$). Similarly, in the mid-range bracket ($1.0 < d \leq 2.5$ km), connected pairs maintain higher correlation than unconnected controls ($\bar{r} = 0.44 \pm 0.18$ vs. $\bar{r}_{\text{null}} = 0.22 \pm 0.14$, $t = 19.8$). We report these comparisons strictly as descriptive sanity checks on internal data consistency rather than formal hypothesis-driven inferences: this

quantitative margin is consistent with physical road corridor continuity beyond isotropic distance decay. Furthermore, users should note that the Gaussian RBF kernel parameter $\sigma = 1.09$ km in Equation (2) is established mathematically from the empirical standard deviation of valid road distances ($\sigma = \text{std}(d_{ij})$) to normalize diffusion weights into $[0, 1]$, rather than serving as a statistical regression curve fit of empirical Pearson correlation values.

Corridor Directionality Consistency with Empirical Flow Measurements. To examine whether directed corridor links (P_f, P_b) and asymmetric edges (1,070 unidirectional and 238 distance-asymmetric pairs) realistically align with urban traffic motion, the derived graph was cross-referenced against longitudinal 1-minute vehicle flow series from the companion pilot study across the identical 608 stations [9] (Zenodo DOI: `https://doi.org/10.5281/zenodo.22089673`). Evaluating the directed cross-correlation function $\text{CCF}_{ij}(\tau) = \text{Corr}(x_i(t), x_j(t + \tau))$ indicates that for valid downstream corridors $(i \to j)$, empirical cross-correlation peaks at positive temporal lags $\tau^* \in [2, 6]$ minutes, consistent with typical metropolitan vehicular transit speeds (20–35 km/h across 1.0–2.5 km corridors), whereas reverse directions exhibit negligible lagged correlation, aligning consistently with expected physical directional transit times. We emphasize that these descriptive characterizations serve as empirical sanity checks verifying the internal physical plausibility and topological sanity of the dataset, rather than asserting predictive benchmark models or inferring causal traffic dynamics.

5. Limitations

1. **Observation Time Window:** The **93.6-hour** observation window captures multi-day diurnal transitions across weekdays and weekends in both the historical metropolitan core and peri-urban expansion districts (Thu Duc City, District 12, Binh Chanh, Hoc Mon, up to 10.99°N), but does not encompass longitudinal multi-month seasonal cycles (monsoon wet vs. dry season).
2. **Snapshot Sampling vs. Continuous Video:** With a 5-minute nominal sampling cadence ($\Delta T = 269.0 \pm 239.7$ s), the dataset is structured for macroscopic traffic state analysis, self-supervised representation pre-training, and spatial graph modeling, but does not support micro-level continuous vehicle trajectory tracking or optical flow.
3. **Surveillance Stream Resolution:** Images are serialized at **512 × 288** JPEG compression as broadcast by the public municipal traffic monitoring service, reflecting real-world public IoT camera stream conditions rather than raw uncompressed sensor feeds. While insufficient for individual vehicle OCR or facial tracking, this resolution is optimal for macroscopic density and self-supervised ViT pre-training while strictly guaranteeing statutory anonymity.
4. **Unlabeled Bulk Corpus:** The 714,123-frame corpus is released unlabeled to facilitate self-supervised pre-training, spatio-temporal graph representation learning, and roadway background extraction without manual annotation biases.
5. **Spatial Representativeness and Network Topology Concentration:** Of the 608 monitored stations, **451 stations (74.18%)** are concentrated across the historic metropolitan urban core and radial avenues (Districts 1, 3, 5, 10, Phu Nhuan, Binh Thanh, and Tan Binh) where vehicular congestion and signalized intersections are densest, while **157 stations (25.82%)** monitor outer ring roads, expressways, and peri-urban gateway corridors (Thu Duc City, District 12, Binh Chanh,

Hoc Mon, extending to 10.9883°N). Consequently, the dataset strongly represents macroscopic arterial corridors and signalized intersections, but does not cover residential alley networks or rural unpaved roads.

6. **Urban Arterial Focus and Modal Routing Generalization:** Camera stations are stationed exclusively on major urban arterials, avenues, and bridge approaches. The road graph routing distances were computed using the standardized motor-vehicle profile in OSRM because both cars and motorcycles share common directional roadways and physical lane corridors along primary arterials. However, this profile does not model dedicated motorcycle-only curbside lanes or informal cut-through routes through residential alleyways (*local residential alleys and side-streets*), which lie outside the coverage of the municipal arterial camera network.
7. **Physical Infrastructure vs. Dynamic Flow Topology:** The derived directed spatial graph models navigable roadway infrastructure connectivity and physical driving distances; it reflects static physical road geometry rather than dynamic, time-varying traffic flow propagation or real-time congestion states.

6. Ethics and Visual Privacy Statement

All visual imagery was recorded from the open public traffic monitoring camera streams broadcast by the Ho Chi Minh City Department of Transportation (`https://giaothong.hochiminhcity.gov.vn`). Established as a public service for civic travel information, congestion monitoring, and flood tracking, these public video streams are made openly accessible for community observation and scientific research under Vietnamese regulations on open government digital data sharing (Decree No. 47/2020/ND-CP). The dataset strictly adheres to personal data protection regulations (Decree No. 13/2023/ND-CP and Law No. 91/2025/QH15). The visual image corpus is released under the Creative Commons Attribution-NonCommercial 4.0 International license (CC BY-NC 4.0) to honor the public civic purpose of the municipal transportation monitoring network, ensuring that traffic surveillance feeds deployed for community flood tracking and congestion awareness remain dedicated exclusively to non-commercial academic research and public educational reuse.

Because cameras are stationed on elevated gantries and masts ($H \geq 6.0$ m) oriented at downward oblique angles and images are broadcast at macroscopic 512×288 resolution, personal identifiable information is optically unresolvable: vehicle registration plates subtend character stroke heights of at most 2.5–2.8 pixels, far below empirical ALPR readability standards ($\geq$ 16–20 pixels [16]), while compulsory helmet and face mask usage eliminates facial biometric features. Across the multi-tiered empirical audit, no readable personal identifiable information (PII) was detected. Given the physical optical resolution constraints (GSD ≥ 1.85 cm/px, stroke height < 3 px), the 95% confidence interval upper bound on residual privacy breach probability is statistically bounded at $p \leq 0.30\%$ on the worst-case sample and $p \leq 0.15\%$ on the stratified random sample via the Hanley and Lippman-Hand Rule of Three [17]. The dataset captures solely macroscopic roadway infrastructure, vehicular flow densities, and environmental traffic states.

Recognizing that Zenodo digital repository archives maintain immutable DOI versioning and open academic licenses cannot be revoked retroactively for previously distributed copies, the authors establish a concrete, enforceable three-tier takedown and retraction protocol:

1. **Official Public Revocation Registry:** Inquiries or redaction requests regarding specific camera viewpoints may be directed to `anhlv.b23kh002@stu.ptit.edu.vn` and `ic4sd-lab@ptit.edu.vn`. Upon receipt and verification within 24 business hours, the affected camera ID is immediately added to an official `REVOKED_STATIONS.csv` registry maintained on the project repository homepage.
2. **Automated Downstream Redaction Utility:** A verified Python utility (`code/redact_revoked_stations.py`) is provided with the repository to enable downstream researchers to programmatically blackout or purge all image files corresponding to revoked station IDs from their local storage mirrors.
3. **Version Deprecation on Data Repositories:** Flagged camera endpoints are immediately removed from the active recording client and lab distribution mirrors, and will be permanently expunged from subsequent minor version releases on Zenodo (Release v2.1+).

7. CRediT Author Statement

Viet-Anh Le: Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Writing – Original Draft, Visualization.
Khanh Nguyen-Trong: Supervision, Conceptualization, Methodology, Writing – Review & Editing, Project Administration.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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